

Submission Summary

Conference Name

2026 IEEE 5th World Conference on Applied Intelligence and Computing

Paper ID

2665

Paper Title

Computational Intelligence for Fetal Health Classification

Abstract

Abstract—Classifying fetal health from cardiotocography (CTG) recordings remains difficult in practice because manual trace interpretation varies widely between clinicians and delays can prove fatal. We report a machine learning pipeline that assigns 2,113 de-duplicated CTG records to Normal, Suspect, or Pathological classes using the original 21 extracted features. Eleven model configurations were benchmarked, covering Gradient Boosting, Random Forest, Extra Trees, Decision Tree, SVM, KNN, Logistic Regression, MLP, and soft-voting ensembles. Controlled ablation experiments show that several popular preprocessing steps—IQR-based outlier capping, log transforms, hand-crafted feature engineering, and SMOTE variants—each hurt the top-performing tree ensembles rather than helping them. Instead, we propose a two-stage strategy: first, cost-aware sample weighting during Gradient Boosting training (weights 1:5:10 for Normal, Suspect, Pathological); second, post-hoc probability threshold adjustment at prediction time. The threshold-tuned model reaches 96.22% accuracy and, more importantly, 100% Pathological recall—no missed pathological case in the test set—at the expense of only three additional false alarms (Pathological precision 92.11%). Suspect recall stands at 81.03%. Five-fold stratified cross-validation on the training partition yields 94.08% $\pm$ 1.17% accuracy and 87.86% $\pm$ 1.75% Pathological recall, indicating stable generalization. Thresholds were selected by minimizing a weighted clinical cost on a held-out validation split, not on the test data, to guard against overfitting. The results argue for metric-driven optimization rather than blanket preprocessing when building classifiers for safety-critical medical applications.

Index Terms—Fetal Health Classification, Cardiotocography, Gradient Boosting, Class Imbalance, Threshold Tuning, Recall Optimization, Machine Learning

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Computational intelligence

Submission Files

Fetal_Health_Classification_Research_Paper.pdf (5.5 Mb, 2026-04-27 16:40:30 GMT+5:30)

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